

4.2 OPTIMIZATION WITH EQUALITY CONSTRAINTS

OPTIMIZATION WITH EQUALITY CONSTRAINTS

■ Example

Consider the following problem:

$$\begin{cases} \max f(x, y) = xy \\ \text{s. a. } x + y = 6 \end{cases}$$

In this case, we could incorporate the constraint in the objective function and transform the optimization problem with an equality constraint into an unconstrained optimization problem.

$$\begin{cases} \max f(x, y) = xy \\ \text{s. a. } \underbrace{x + y = 6}_{y = 6 - x} \end{cases} \Leftrightarrow \max \underbrace{x(6 - x)}_{f(x, 6 - x)} = 6x - x^2$$

$$f'(x) = 0 \quad (\Rightarrow) \quad 6 - 2x = 0 \quad (\Rightarrow) \quad x = 3$$

$$f''(x) = -2 \quad \Rightarrow \quad f''(3) = -2 < 0. \quad \text{So, } f(3, 3) = 9 \text{ is a } \underline{\underline{\text{maximum}}}$$

OPTIMIZATION WITH EQUALITY CONSTRAINTS

■ Example (cont.)

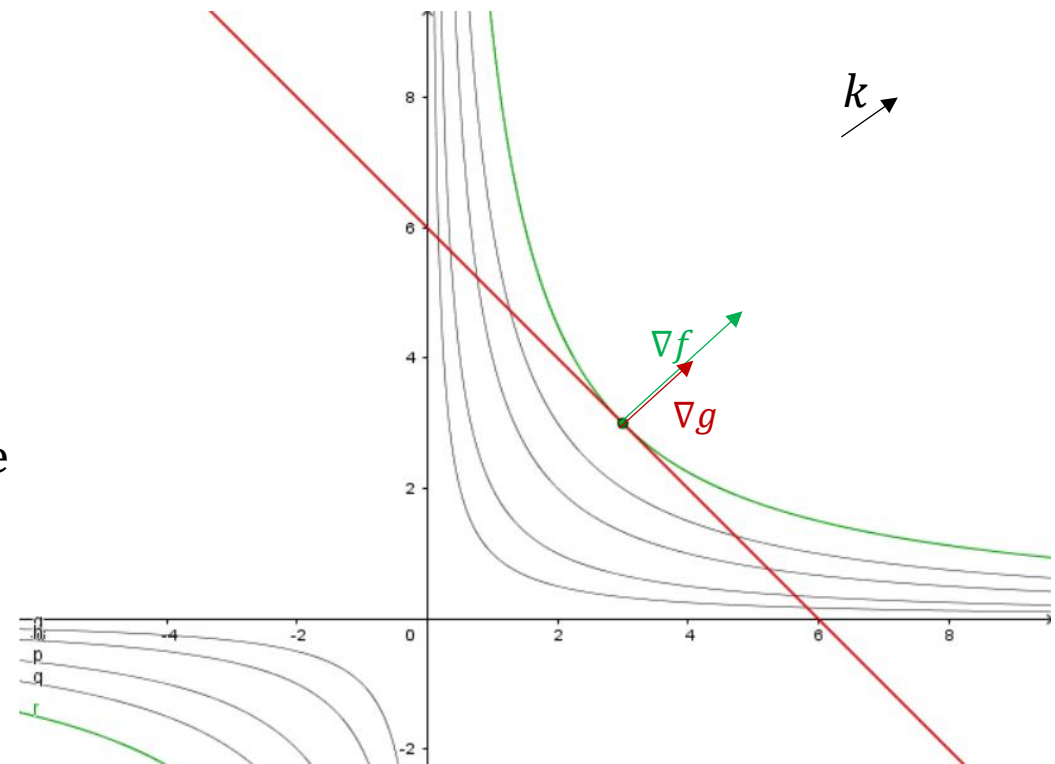
Let's now consider a graphical approach to solve the same

problem
$$\begin{cases} \max f(x, y) = xy \\ \text{subject to } x + y = 6 \\ \quad \quad \quad \underline{g(x, y)} = 6 \end{cases}$$

- Represent graphically the constraint: $x + y = 6$.
- Draw level curves of the objective function:

$$f(x, y) = k \Leftrightarrow xy = k$$
- The maximum is at the tangent point between a level curve and the constraint.
- How to compute this tangent point?

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 6 \end{cases}$$



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■ Example (cont.)

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 6 \end{cases} \quad (=\Rightarrow)$$

$$\begin{aligned} \Rightarrow \begin{cases} (y, x) = \lambda (1, 1) \\ x + y = 6 \end{cases} & \quad (\Rightarrow) \quad \begin{cases} y = \lambda \\ x = \lambda \\ \lambda + \lambda = 6 \end{cases} \quad (\Rightarrow) \quad \begin{cases} y = 3 \\ x = 3 \\ \lambda = 3 \end{cases} \end{aligned}$$

So, $f(3, 3) = 9$ is the maximum value of f when subject to the constraint $x + y = 6$

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Lagrange Multipliers Method** (n=2 variables, p=1 constraint)

Consider the problem
$$\begin{cases} \max f(x, y) \text{ or } \min f(x, y) \\ \text{subject to } g(x, y) = b \end{cases}$$

where $f, g: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be functions of class C^2 in an open set D such that $\nabla g(x, y) \neq \mathbf{0}$ at any point of the constraint set.

Let's define an auxiliary function, the **Lagrangian Function** (λ is called Lagrange multiplier and we will see later the interpretation of this parameter):

$$L(\lambda, x, y) = f(x, y) + \lambda(b - g(x, y))$$

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Lagrange Multipliers Method** (n=2 variables, p=1 constraint)

First-order Conditions: By solving the following system, we find the critical points

$$\nabla L(\lambda, x, y) = \mathbf{0} \Leftrightarrow \begin{cases} \frac{\partial L}{\partial \lambda} = 0 \\ \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \end{cases}$$

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Lagrange Multipliers Method** (n=2 variables, p=1 constraint)

Second-order Conditions: Consider the hessian matrix of the Lagrange function (**Bordered Hessian matrix**)

$$\bar{H}(\lambda, x, y) = \begin{bmatrix} \frac{\partial^2 L}{\partial \lambda^2} & \frac{\partial^2 L}{\partial \lambda \partial x} & \frac{\partial^2 L}{\partial \lambda \partial y} \\ \frac{\partial^2 L}{\partial x \partial \lambda} & \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial^2 L}{\partial y \partial \lambda} & \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{\partial^2 L}{\partial \lambda \partial x} & \frac{\partial^2 L}{\partial \lambda \partial y} \\ \frac{\partial^2 L}{\partial x \partial \lambda} & \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial^2 L}{\partial y \partial \lambda} & \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{bmatrix}$$

Analyse the last $(n - p)$ principal minors of the bordered hessian matrix at the critical points. In this particular case it means that we only have to analyze the last $(2-1=1)$ principal minor:

If $|\bar{H}| > 0 \Rightarrow$ local maximum
 If $|\bar{H}| < 0 \Rightarrow$ local minimum

OPTIMIZATION WITH EQUALITY CONSTRAINTS

■ Example

Use Lagrange Multipliers Method to solve the initial problem $\begin{cases} \max f(x, y) = xy \\ \text{subject to } \underbrace{x + y}_{g(x, y)} = \underbrace{6}_b \end{cases}$

$$L(\lambda, x, y) = f(x, y) + \lambda(b - g(x, y))$$

$$L(\lambda, x, y) = xy + \lambda(6 - x - y)$$

First-Order conditions: $\nabla L(\lambda, x, y) = (0, 0, 0) \Leftrightarrow$

$$\begin{cases} \frac{\partial L}{\partial \lambda} = 0 \\ \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 6 - x - y = 0 \\ y - \lambda = 0 \\ x - \lambda = 0 \end{cases} \Leftrightarrow \begin{cases} 6 - \lambda - \lambda = 0 \\ y = \lambda \\ x = \lambda \end{cases} \Leftrightarrow \begin{cases} \lambda = 3 \\ y = 3 \\ x = 3 \end{cases}$$

Second-order conditions: study the sign of the last $(2-1=1)$ principal minor of the bordered Hessian matrix at the critical point, $\bar{H}(3, 3, 3)$.

$$\bar{H}(\lambda, x, y) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \quad (= \bar{H}(3, 3, 3))$$

$$|\bar{H}(3, 3, 3)| = \begin{vmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = 0 + 1 + 1 - 0 - 0 - 0 = 2 > 0$$

As $m=2$ and $p=1$ and the relevant principal minor is positive, we conclude that $f(3, 3) = 9$ is maximum of f under the given constraint

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■ Exercise

Solve the problem $\begin{cases} \text{optimize } f(x, y) = x^2 + y \\ \text{subj. to } \underbrace{x + y = 2}_{g(x, y)} \end{cases}$

- 1) Using a graphical approach.
- 2) Using Lagrange Multipliers Method.

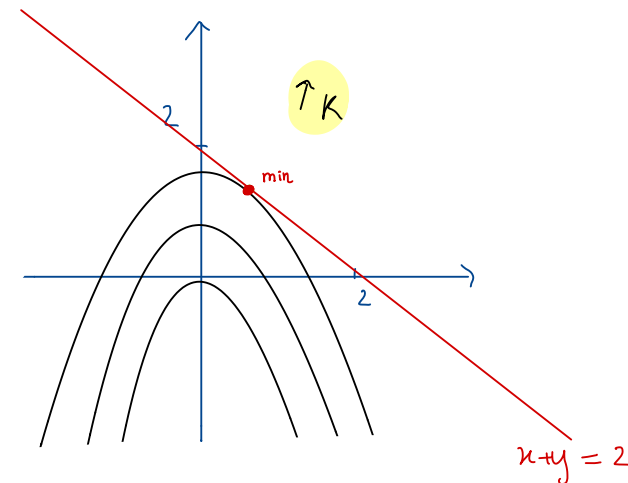
Using a graphical approach:

(1) Draw the constraint: $x + y = 2$

(2) Draw level sets of f : $x^2 + y = K \Leftrightarrow$

$$\Leftrightarrow y = -x^2 + K$$

We are looking for the tangency point between one of the level curves and the constraint.



It is a minimum because it corresponds to the "first" level curve (lowest value of K) that intersects the constraint.

(3) Calculus of the solution:

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 2 \end{cases} \Leftrightarrow \begin{cases} (2x, 1) = \lambda(1, 1) \\ x + y = 2 \end{cases} \Leftrightarrow \begin{cases} 2x = \lambda \\ 1 = \lambda \end{cases} \Leftrightarrow \begin{cases} x = \frac{1}{2} \\ \lambda = 1 \end{cases} \Leftrightarrow y = 2 - \frac{1}{2} \Rightarrow \boxed{y = \frac{3}{2}}$$

So, $f\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{1}{4} + \frac{3}{2} = \boxed{\frac{7}{4}}$ is the minimum of f when subject to the constraint $x + y = 2$.

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■ Exercise (cont.)

Using Lagrange Multipliers Method:

$$L(\lambda, x, y) = x^2 + y + \lambda(2 - x - y)$$

First-Order Conditions: $\nabla L(\lambda, x, y) = (0, 0, 0) \Rightarrow$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial \lambda} = 0 \\ \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 2 - x - y = 0 \\ 2x - \lambda = 0 \\ 1 - \lambda = 0 \end{cases} \Leftrightarrow \begin{cases} y = 2 - \frac{1}{2} \Rightarrow \boxed{y = \frac{3}{2}} \\ \boxed{x = \frac{1}{2}} \\ \boxed{\lambda = 1} \end{cases}$$

Second-order Conditions: we must study the sign of the last $(2-1=1)$ principal minor of $\bar{H}(1, \frac{1}{2}, \frac{3}{2})$

$$\bar{H}(\lambda, x, y) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad (= \bar{H}(1, \frac{1}{2}, \frac{3}{2}))$$

$$|\bar{H}(1, \frac{1}{2}, \frac{3}{2})| = \begin{vmatrix} 0 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 0 \end{vmatrix} = -2 < 0$$

As the problem has 2 variables and 1 const. and the relevant principal minor is negative we conclude that $f(\frac{1}{2}, \frac{3}{2}) = \boxed{\frac{7}{4}}$ is the minimum of f when subject to the constraint $x + y = 2$.

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Lagrange Multipliers Method** (n variables, p constraints)

Consider the problem:

$$\begin{array}{ll} \max f(x_1, \dots, x_n) & \text{or } \min f(x_1, \dots, x_n) \\ \text{subject to} & \begin{cases} g_1(x_1, \dots, x_n) = b_1 \\ \dots \\ g_p(x_1, \dots, x_n) = b_p \end{cases} \end{array}$$

where $f, g_1, \dots, g_p: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be functions of class C^1 in an open set D and ∇g_j are linearly independent.

Let's define an auxiliar function, the **Lagrange Function**:

$$L(\lambda_1, \dots, \lambda_p, x_1, \dots, x_n) = f(x_1, \dots, x_n) + \lambda_1(b_1 - g_1(x_1, \dots, x_n)) + \dots + \lambda_p(b_p - g_p(x_1, \dots, x_n))$$

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Lagrange Multipliers Method** (n variables, p constraint)

First-order Conditions

$$\nabla L(\lambda_1, \dots, \lambda_p, x_1, \dots, x_n) = \mathbf{0} \Leftrightarrow \begin{cases} \frac{\partial \mathcal{L}}{\partial \lambda_j} = 0 & (j = 1, \dots, p) \\ \frac{\partial \mathcal{L}}{\partial x_i} = 0 & (i = 1, \dots, n) \end{cases}$$

Second-order Conditions

Analyse the last $(n - p)$ principal minors of the bordered hessian matrix (**relevant principal minors**) at the critical points found in the first-order conditions.

If the relevant principal minors alternate in sign starting with the sign of $(-1)^{p+1} \Rightarrow$ Maximum

If the relevant principal minors have all the same sign of $(-1)^p \Rightarrow$ Minimum

OPTIMIZATION WITH EQUALITY CONSTRAINTS

■ Example

$$n=3 \text{ and } p=1$$

Find the extremals of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $x - y + z = 1$

$$L(\lambda, x, y, z) = x^2 + y^2 + z^2 + \lambda(1 - x + y - z)$$

First-order Conditions: $\nabla L(\lambda, x, y, z) = (0, 0, 0, 0) \Leftrightarrow$

$$\Leftrightarrow \begin{cases} L_\lambda = 0 \\ L_x = 0 \\ L_y = 0 \\ L_z = 0 \end{cases} \Leftrightarrow \begin{cases} 1 - x + y - z = 0 \\ 2x - \lambda = 0 \\ 2y + \lambda = 0 \\ 2z - \lambda = 0 \end{cases} \Leftrightarrow \begin{cases} 1 - \frac{\lambda}{2} - \frac{\lambda}{2} - \frac{\lambda}{2} = 0 \\ x = \frac{\lambda}{2} \\ y = -\frac{\lambda}{2} \\ z = \frac{\lambda}{2} \end{cases} \Leftrightarrow$$

$$\begin{cases} \frac{3\lambda}{2} = 1 \Leftrightarrow \lambda = \frac{2}{3} \end{cases}$$

$$\Leftrightarrow \begin{cases} x = \frac{1}{3} \\ y = -\frac{1}{3} \\ z = \frac{1}{3} \end{cases}$$

Second-order Conditions: Study the sign of the last $3-1=2$ principal minors of $\bar{H}(\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3})$

$$\bar{H}(\lambda, x, y, z) = \begin{bmatrix} 0 & -1 & 1 & -1 \\ -1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{bmatrix} \quad \left(= \bar{H}(\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}) \right)$$

The relevant principal minors are:

$$|\bar{H}_3| = \begin{vmatrix} 0 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -2 - 2 = \textcircled{-4}$$

$$\begin{matrix} 0 & -1 & 1 \\ -1 & 2 & 0 \end{matrix}$$

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Laplace along last row

$$|\bar{H}_4| = \left| \bar{H}\left(\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right) \right| = \begin{vmatrix} 0 & -1 & 1 & -1 \\ -1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{vmatrix} = (-1)(-1)^5 \begin{vmatrix} -1 & 1 & -1 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & -1 \\ 2 & 0 & 0 \end{vmatrix} + 2 \times (-1)^8 \begin{vmatrix} 0 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} =$$

$$= -4 + 2 \times (-4) = \boxed{-12}$$

As the relevant principal minors are all of the same sign as $(-1)^7$ then $f\left(\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3}$ is the minimum of f under the given constraint.

OPTIMIZATION WITH EQUALITY CONSTRAINTS

- **Interpretation of the Lagrange Multiplier**

Considering the problem:

$$\begin{cases} \text{Optimize } f(x, y) \\ \text{subj. to } g(x, y) = b \end{cases}$$

The Lagrange multiplier λ gives approximately the change in the value of the optimal of the function when the constant b of the constraint increases by 1 unit.